

Data Visualisation Using Tableau and Power BI

Data visualisation plays a vital role in transforming raw data into meaningful insights. It helps reveal patterns, trends and anomalies that might otherwise remain hidden in spreadsheets or tables. Whether the goal is to explore data for insights or to present findings clearly to others, visualisation bridges the gap between analysis and communication.

Two of the most widely used tools for creating such visualisations are Tableau and Power BI. Both tools offer powerful features that allow users to connect to data sets, model and clean data and design interactive dashboards. Tableau is known for its intuitive drag-and-drop interface and visual flexibility, while Power BI offers tight integration with the Microsoft ecosystem and robust modelling capabilities through DAX and Power Query.

Understanding the differences and strengths of each tool is important for choosing the right approach depending on the analytical need. Throughout this note, learners will explore how Tableau and Power BI support both exploratory and explanatory analysis, from creating basic charts to designing interactive dashboards.

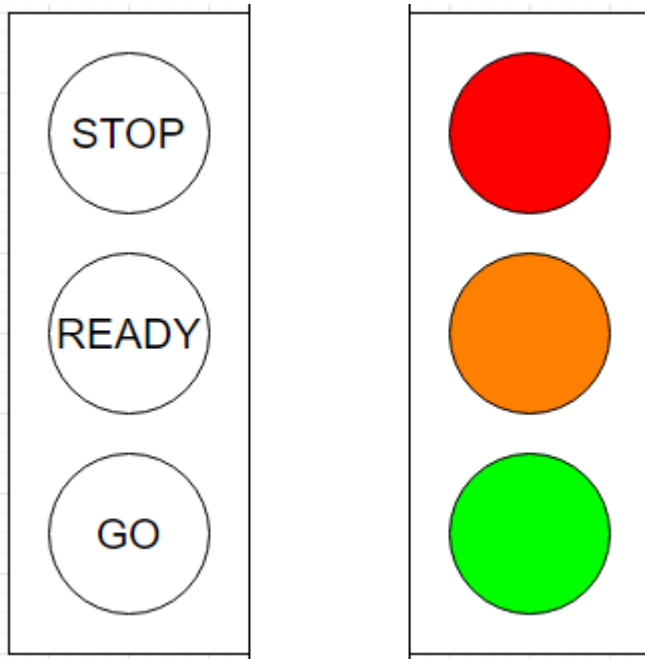
The focus throughout the module remains on building practical skills like connecting to data, applying transformations, creating calculated fields and assembling visual elements into meaningful stories. By mastering these tools, one can turn complex data sets into clear and actionable insights that support informed decision-making.

Introduction to Tableau

Data analytics helps uncover key business insights, and data visualisation plays a vital role in making those insights clear and accessible. Tableau is widely used for creating such visualisations due to its ease of use and powerful features. The key topics covered in the module are about the role of visualisation in analytics and why Tableau stands out among tools like Excel and Power BI. We also examine two key forms of analysis—exploratory and explanatory—and learn how Tableau supports both. We then are introduced to Tableau's interface, and methods to connect data in various formats, and essential functions for creating charts and graphs using numeric, string, logical and date fields.

Why Data Visualisation?

Reading through large volumes of raw data can be time consuming and often leads to missed insights. Data visualisation helps simplify this process by representing complex information in a graphical format, making it easier to identify patterns, trends and outliers. It not only enhances one's understanding but also allows for quickly interpreting and communicating results.



For example, the coloured lights used in traffic signals can be seen clearly from a distance and are easy to interpret. Imagine if traffic signals suddenly had words instead of colours. This would create a great magnitude of chaos. The coloured lights make traffic management easier and simple for people to follow traffic signals, even if they do not know to read and write.

That is where visualisation plays an important role in transforming information into simple graphs, involving the usage of colours, shapes, sizes and more, so that users can ingest the information without the need to read and remember lengthy texts and numbers.

Parameters of Visualisation

The four important parameters of visualisation are as follows:

- **Colour:** Colours help the user differentiate between various categories of data.
- **Shape:** Shapes also help the user differentiate between various categories of data.
- **Size:** The size of different bubbles or bars in a chart helps the user compare the values of each category.
- **Proximity:** Proximity refers to the closeness of bubbles or bars. Having similar categories closer to each other helps cluster them for further analysis.

What is Tableau?

Tableau is a widely used business intelligence tool developed in 2003. It is trusted by major organisations like EY, Facebook and Deloitte for visualising data and deriving insights. Known for its ease of use and strong visual capabilities, Tableau holds a leading position in the analytics industry compared to tools like Power BI, QlikView and Spotfire.

It helps convert raw data into structured information to support effective decision-making. Two versions of Tableau are available—Tableau Public, which is free with some limitations, and Tableau Desktop, which offers full functionality for a 14-day trial. Either version can be used to create visualisations, depending on the duration and scope of use.

Why Tableau?

Tableau has become one of the most preferred tools in data visualisation due to its simplicity, flexibility and powerful features. With an intuitive drag-and-drop interface, it allows users to build interactive dashboards and visualisations without the need for coding. Its ability to support real-time sharing, handle a wide range of data formats and connect with various databases makes it a versatile choice for business intelligence. Tableau offers multiple products tailored to different needs, enabling individuals and organisations to analyse and present data effectively across various platforms.

According to a report by Gartner, Tableau is used by over 60% of the companies in the world. Tableau has several key features and functionalities. Some of them are as follows:

- It is easy to learn because of its simple drag-and-drop interface; there is almost no coding required.
- It has interactive dashboards.
- Visualisations and dashboards can be shared with different stakeholders in real time as per requirements.
- It offers cloud support as well.

Tableau supports data in several formats. Some of these data formats are as follows:

- PDF
- JSON
- Text
- Excel

Even the data stored in different types of databases can be integrated into Tableau. Some of these databases are as follows:

- MySQL
- Salesforce
- MongoDB
- Amazon Redshift
- Oracle

Tableau offers many products that are designed to suit various analytical and visualisation needs. Some of these products are as follows:

- **Tableau Desktop:** This is a licensed software used for creating charts and graphs.

- **Tableau Public:** This is similar to Tableau Desktop. However, Tableau Public is free and offers fewer features than Tableau Desktop.
- **Tableau Online:** This is used for sharing visualisations and dashboards. For Tableau Online, the dashboards and reports are stored in the cloud, and you need to pay for these services.
- **Tableau Server:** This is similar to Tableau Online. However, the dashboards and reports are stored in on-premise spaces in this case.
- **Tableau Reader:** This is similar to Acrobat Reader, which is used for viewing text documents and reading/viewing dashboards. Tableau Reader comes in handy for users who do not have Tableau installed on their systems. They can still view the dashboards but cannot edit them.

Difference Between Tableau and Excel

The differences between Tableau and Excel are summarised in the following table.

Tableau	Excel
It is a data visualisation tool.	It is a spreadsheet that is generally used for storing data.
It provides higher performance than Excel.	It provides lower performance as compared to Tableau.
It enables the creation of a lot of custom charts.	It has a limited number of charts available.
It takes less time to create dashboards without a lot of coding.	It requires more time to create dashboards and may involve more coding.
It can be easily integrated with R and Python.	It cannot be integrated with R and Python.
There is no limit on the amount of data that can be handled in Tableau.	A maximum of one million rows of data can be handled in Excel.

Difference Between Tableau and Power BI

The differences between Tableau and Power BI are summarised in the table given below.

Tableau	Power BI
It is primarily a business intelligence tool used for creating reports.	It is primarily a data analytics tool that allows you to perform Extract Transform Load (ETL) tasks and text mining.
It can be connected to virtually all types of data sources and servers.	It has access to limited types of data sources and servers.
There is no limit on the amount of data that can be handled in Tableau.	A maximum of 10 GB of data can be handled in each workspace.
It provides higher performance.	It provides lower performance as compared to Tableau.
It is quite expensive to get a one-year license.	It is much cheaper than Tableau.

So, Tableau has some clear advantages over both Excel and Power BI. The only major drawback is that it is expensive. It is almost time for you to familiarise yourself with the Tableau interface. However, an analyst needs to understand the difference between exploratory and explanatory analysis before performing any analytics.

Exploratory vs Explanatory Analysis

The meanings of the terms ‘exploratory analysis’ and ‘explanatory analysis’ are quite clear. While the term ‘exploratory’ is an extension of ‘explore’, the term ‘explanatory’ is an extension of ‘explain’. The difference between the two types of analyses is as below:

Exploratory analysis:

Exploratory analysis is often the first step of data analysis. Here we get familiar with data, ask questions, visualise the data in a number of forms, look for relationships between the variables, and look for outliers, patterns and trends in data. The output of exploratory analysis is usually only for us, the data analysts.

Explanatory analysis:

Explanatory analysis is what happens when we identify some interesting observations in the data. We then create visuals to present our findings. The output of an explanatory analysis is generally for the public.

Exploratory analysis is often the first step of data analysis. Some of the steps involved in performing an exploratory analysis are as follows:

- Getting familiar with the data
- Asking questions
- Visualising the data in several forms
- Looking for relationships between variables
- Looking for outliers in the data
- Identifying any patterns and trends that emerge from the data

On the other hand, explanatory analysis involves explaining some interesting observations that you may have made after performing exploratory analysis. In this analysis, you create visuals to present your findings.

The output of exploratory analysis is usually meant only for data analysts. However, the output of an explanatory analysis is for the general public.

Visualising and Analysing Data with Tableau

Visualisation is at the core of any analysis in Tableau. This section focusses on some of the most commonly used chart types that help uncover patterns and insights in a given data set. Bar charts, line charts, area charts, box plots, pie charts and tree maps each serve specific purposes depending on the nature of the data and the type of insight being sought. Understanding the structure and application of these charts is essential for selecting the right visual to communicate your findings effectively.

In addition to visual elements, Tableau provides key features such as filters, hierarchies, and grouping, which allow for more refined and layered data explorations. These tools enable the user to focus on relevant segments, organise data meaningfully and simplify complex views. Another important concept is calculated fields, which help derive new variables from existing data and support deeper analysis by introducing customised logic and computations.

Bar Charts

Bar charts are generally used to compare data values or trace changes in data over a period. In Tableau, various operations can be performed using bar charts:

- Sorting data values in ascending and descending order in a graph
- Swapping the X and Y axes in a graph
- Changing the colour of a graph
- Adding labels to the values in a graph
- Creating a calculated field that is obtained by performing calculations on two or more existing fields in a data set

Line Charts and Filters

Line charts are used to analyse data values over a period. Along with this, filters are applied to narrow down the analysis:

- Filters are used to view only the required categories from an attribute. For example, you can create and use a filter to analyse data about men or people of a certain age group.
- Tooltips can be customised such that you can view them easily in your visualisations.

Area Charts

An area chart is a line chart where the area between the line and the axis is shaded with a certain colour. Typically, these charts are used to represent the accumulated total values over a period and are the conventional way of displaying stacked lines.

Box Plots

Box plots are quite effective when you want to compare the distribution of two quantitative variables. The constituents of a box plot are as follows:

- The line inside the box denotes the median value, and 50% of the data lies above that value, while 50% lies below it.
- The boundaries of the box denote the 25th and the 75th percentiles.

- The whiskers denote the lowest and the highest values in the data set (excluding outliers).
- Any observations lying below or above the whiskers are considered to be outliers. Outliers are the data points that have absurdly higher or lower values than other observations.
- Pivoting is the process of converting rows into columns in a data set and vice versa.
- You can format data if it contains any extraneous characters, such as spaces.

Maps

Maps is an online functionality provided in Tableau. To build a simple map, your data source must contain location data (location names or latitude and longitude coordinates). The map that contains points in each state, for example, is called a point map. The map that contains states filled with varying colours, for example, is called a filled map or a polygon map.

Hierarchies

A hierarchy of features, such as country, state and city, can be created to obtain richer visualisations using the same data set.

Pie Charts

Pie charts are effective tools for visualising a particular category as a ratio of the total available categories.

Tree Maps

- Tree maps plot the quantitative variables for each category of a dimension.
- The values of the quantitative variables are represented by the size and the colour intensity of the plot.
- They can accommodate only two measures: one to control the size and the other to control the colour. However, they can accommodate any number of dimensions, as it would only increase the number of rectangles on the tree map.

You also learnt that grouping is an operation carried out in Tableau to group values of certain categories together. This is usually done to remove data entry errors.

Dashboards

A dashboard is a collection of multiple sheets arranged in a visually appealing manner. They allow you to visualise and analyse different types of data simultaneously.

Calculated Fields

Calculated fields in Tableau allow you to create new data columns by applying custom formulas to existing fields. They help derive insights, perform computations and customise data analysis within visualisations.

TWB vs TWBX

You can save your Tableau workbook in two formats: .twb and .twbx. The table below will help you understand the differences between the two formats and the advantages of one file format over the other.

.twb	.twbx
Only the visualisations are stored, not the data source.	Both the data source and the visualisations are stored.
It has a comparatively small size, so it is easier to share the file.	It has a comparatively large size, so it is relatively difficult to share the file.

Introduction to Power BI

Power BI is a business intelligence tool developed by Microsoft that enables users to connect to various data sources, transform data and create interactive reports and dashboards. It is widely used across industries for tasks such as sales tracking, financial reporting, customer analysis and operational monitoring. Power BI supports informed, data-driven decision-making by providing real-time insights in a visual format.

The Power BI ecosystem consists of three key components—Power BI Desktop, Power BI Service and Power BI Mobile—which together allow users to create, publish and share reports. Reports can be hosted on the cloud through Power BI Service or on-premises using Power BI Report Server. The platform also includes AI-powered features such as Power BI Copilot, which assists in building visuals and analysing data, provided certain licensing and setup requirements are met.

Data can be imported into Power BI from various sources such as Excel, SQL Server and cloud platforms like Azure and Salesforce. Power BI supports both Import and DirectQuery modes, allowing flexibility based on performance and use case. Data transformation tools such as column splitting and query merging help prepare data for analysis, while data modelling features allow users to build relationships between tables for deeper insights.

The Power BI interface is designed for ease of use, with dedicated sections for data, models and visual design. As we explore Power BI further, attention will be given to the full process—from data ingestion and transformation to report creation and sharing—so meaningful insights can be derived effectively.

Microsoft Power BI, a powerful and user-friendly tool for transforming raw data into interactive dashboards and reports. Designed for business users, it supports data connection, transformation, modelling, and visualisation—all within an intuitive interface. With seamless integration across platforms like Excel, SQL Server and cloud services, Power BI serves as a versatile solution for organisations of all sizes.

Key Features of Power BI

Power BI offers a range of features that make it a powerful and accessible tool for data analysis and visualisation.

- **User-Friendly Interface**

The drag-and-drop functionality allows users to build interactive reports with ease, without requiring any coding knowledge.

- **Diverse Visual Options**

A wide variety of visuals such as bar charts, maps and custom visuals are available to represent data effectively based on the context and need.

- **Interactive Reports**

Reports in Power BI support features like filtering and drill down, enabling users to explore data points in detail and uncover deeper insights.

- **Real-Time Updates**

Power BI can connect to live data sources including Excel files, SQL databases and cloud services, ensuring that reports remain current and accurate.

- **Easy Sharing and Collaboration**

Reports and dashboards can be published via Power BI Service, allowing seamless sharing and collaboration within teams and across organisations.

Power BI Ecosystem

Microsoft Power BI is a robust data-reporting platform that provides a suite of components and functionalities, making it highly beneficial for data analysis and reporting.

- **Power BI Desktop** allows you to connect to various data sources, clean and organise data, and create visual reports and dashboards to gain insights into key business metrics, trends and performance.

The key activities carried out in Power BI Desktop are as follows:

- Connecting with your data
- Data cleaning and transformation
- Creating data models

- Creating reports and dashboards

- **Power BI Service**

This is a cloud platform for sharing and collaborating on reports with features like scheduled data refreshes and Q&A for instant insights.

- **Power BI Mobile**

It allows you to access reports and dashboards on the go, ideal for real-time insights.

- **Power BI Report Server**

This is an on-premises solution for hosting and managing reports within an organisation's infrastructure, ensuring data security and compliance.

The various components of Power BI integrate to support a seamless workflow as follows:

- **Report creation:** Connect to data sources, transform data and design reports using Power BI Desktop
- **Publishing:** Upload completed reports to Power BI Service for cloud hosting and sharing
- **Sharing:** Securely share reports with stakeholders for collaborative analysis
- **Mobile access:** Stakeholders access reports on the go through Power BI Mobile apps
- **Data refresh:** Schedule automatic data refreshes to keep reports updated with the latest information

Understanding each Power BI component allows organisations to leverage the platform's full potential, driving data-driven transformation. Its integration with the Microsoft ecosystem enables seamless data import from Excel, advanced analytics via Azure and collaboration through Microsoft Teams, streamlining workflows and maximising existing investments.

After understanding the key components and core tools of Power BI, it is important to explore the different licensing options available. Power BI offers Free, Pro and Premium plans, each designed to suit varying organisational needs and budgets. In addition to the cloud-based services, Power BI provides an on-premises solution through Power BI Report Server. Knowing the differences between these options helps select the most appropriate setup for report creation, sharing and collaboration based on the scale and requirements of the organisation.

Following are the Power BI's licensing options:

- **Power BI Free:** Entry-level option for individuals to create and publish reports in a personal workspace, with limited sharing and collaboration features
- **Power BI Pro:** Enables collaboration, report sharing and advanced analytics, ideal for teams and organisations
- **Power BI Premium:** Enterprise-level solution with dedicated capacity, offering advanced features like paginated reports, AI insights and scalable options for large-scale deployments

The table below compares features across different Power BI licensing options:

Feature	Power BI Free	Power BI Pro	Power BI Premium
Report Creation	Yes	Yes	Yes
Personal Workspace	Yes	Yes	Yes
Sharing & Collaboration	Limited	Yes	Yes
App Workspaces	No	Yes	Yes
Embedded Analytics	No	Yes	Yes
Data Refresh	Limited	Yes	Enhanced
Paginated Reports	No	No	Yes
AI Insights	Limited	Limited	Enhanced
Dedicated Capacity	No	No	Yes
On-Premises Option	No	No	With Report Server

Let's further expand our understanding by exploring **Copilot for Microsoft Fabric**. This integration of generative AI within Power BI empowers users to uncover deeper insights and craft more impactful, data-driven reports.

As we have now reviewed integrating Copilot in Power BI, we will discuss the key features and requirements to utilise it.

Copilot Features:

- Suggests measures for insights
- Helps with DAX formulas
- Filters narrative visuals
- Auto-generates Q&A synonyms
- Recommends report content
- Creates summary reports with text

Requirements:

- **Admin setup:** The organisation's admin must activate Copilot in Microsoft Fabric Admin Portal.
- **License:** This needs Microsoft Fabric (F64 or higher) or Power BI Premium (P1 or higher).

Power BI Installation

While understanding Power BI is important, being able to install and use it effectively is crucial for hands-on application. Below are two ways to install Power BI Desktop, along with a link that provides details on the necessary system requirements.

Method 1: Downloading from the Official Microsoft Website

1. Visit the Power BI Desktop download page.
2. Select your language and click 'Download'.
3. Choose the 64-bit or 32-bit version depending on your system.
4. Run the installer and follow the setup wizard.
5. Launch Power BI Desktop from the Start menu.

Method 2: Installing via Microsoft Store

1. Open Microsoft Store and search for 'Power BI Desktop'.
2. Click 'Get' to install.
3. Launch Power BI Desktop from the Store or Start menu.

Installing Power BI on Mac Using Parallels

1. Install Parallels Desktop on your Mac using this [link](#).
2. Set up a Windows environment in Parallels.
3. Download and install Power BI Desktop from the official website.
4. Use Power BI within Parallels.
5. For a detailed video tutorial on how to set up Parallels and install Power BI, check out this [link](#).

To access Power BI's full set of features, registration must be completed using a work or school email address. Personal email domains such as @gmail.com are not supported for full access, and without proper registration, key functionalities such as report sharing will be restricted. After completing the sign-up, it is also possible to invite external users, provided the necessary admin permissions are enabled.

The steps to register for a free Power BI license are as follows:

1. Go to this link.
2. Enter your email address and then select sign up.
3. Verify your account. Check your email address for your verification code. On the next screen, enter your information and the verification code from the email.
4. Select a region, review the policies that are linked from the screen, then select 'Get Started'.
5. You can now begin using Power BI Desktop after signing in.

Logging into Power BI with a Personal Account

You can also log into Power BI using a personal Microsoft account. Here is how:

1. **Open Power BI website:** Open your browser and visit the official Power BI website.
2. **Sign up:** Click 'Start free' and sign up with a personal Microsoft account (Outlook, Hotmail or Live). If you do not have one, you can create it during the process.
3. **Complete signup:** Verify your email and follow the instructions to finish.
4. **Access Power BI:** Once logged in, you can start using Power BI's basic features. For more details, visit [this](#) link.

Power BI Interface

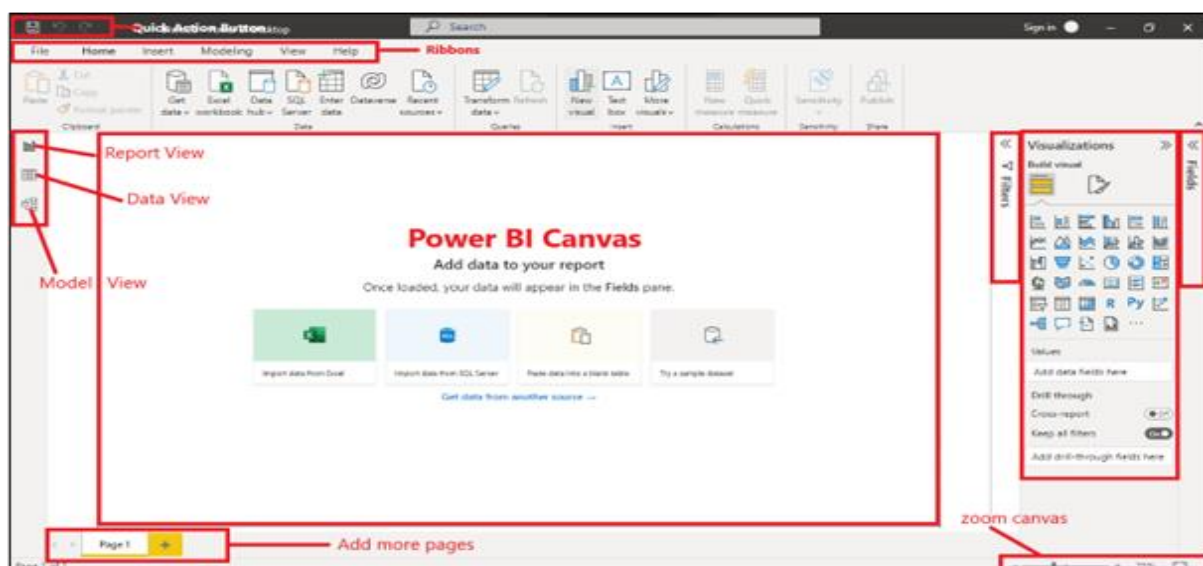
Power BI Desktop has a user-friendly interface that makes it easier to turn data into engaging visuals. By getting to know its main parts, you can navigate the platform more smoothly and make the most of its features.

Understanding the Power BI Desktop interface is essential for creating effective reports and visualisations. Here's a simplified overview of its main components.

1. **Ribbon:** Located at the top, the ribbon organises commands into tabs:
 - **Home:** For connecting to data, transforming it, and managing reports.
 - **Insert:** To add visuals, text boxes, buttons, and other elements.
 - **Modelling:** For creating relationships, calculated columns, and measures.
 - **View:** To adjust the appearance of your report canvas and panes.
 - **File and Help:** For additional functionalities and support.
2. **Data Pane:** On the right side, this pane lists all tables and columns in your data model. You can drag fields from here onto the report canvas to create visuals.

3. **Visualisations Pane:** Also on the right, it offers a variety of chart types like bar charts, line charts and maps. You can select and customise these visualisations to represent your data effectively.
4. **Filters Pane:** This pane allows you to focus on specific subsets of data, enabling interactive exploration and detailed analysis.
5. **Report Canvas:** The central workspace where you design and arrange your visualisations, bringing your data stories to life.
6. **Page Tabs:** Located at the bottom, these tabs let you create multiple pages within a single report, helping organise your visualisations into sections. Power BI offers customisable PC layouts for detailed reports and advanced analytics, while the mobile layout ensures responsive, touch-friendly access to data on the go.
7. **Views:**
 - **Report View:** The default view for creating and viewing reports
 - **Data View:** Displays your data set in a tabular format for inspection and minor edits.
 - **Model View:** Shows relationships among tables in your data model.
 - **DAX Query View:** Allows you to write and execute DAX queries for data analysis and custom calculations.

Here is a screenshot of the Power BI interface to help you identify and familiarise yourself with its key components as you begin working with the tool.



A standard workflow in Power BI involves several key steps:

1. **Connect to data sources:** Begin by using the 'Get Data' option on the Home tab to connect to various data sources, such as databases, Excel files or online services.
2. **Transform and clean data:** Utilise the Power Query Editor to clean and transform your data, ensuring it is structured appropriately for analysis.
3. **Model data:** In the Modelling tab, establish relationships between tables and create calculated measures using Data Analysis Expressions (DAX) to enhance your data model.

4. **Create visualisations:** Drag fields from the Fields pane onto the report canvas and select suitable visualisations from the visualisations pane to represent your data effectively.
5. **Format visualisations:** Customise your visuals by adjusting colours, fonts, labels and other properties to improve clarity and aesthetics.
6. **Add interactivity:** Incorporate filters, slicers and other interactive elements to allow dynamic exploration of your reports.
7. **Organise report pages:** Structure your visualisations across multiple pages if needed, using the Page tabs at the bottom of the report canvas for better organisation.
8. **Publish and share:** Once your report is complete, publish it to the Power BI Service to share with colleagues or stakeholders.
9. **Customise workspace:** Use the View tab to personalise your workspace, showing or hiding panes and adjusting the layout to suit your preferences.

Next, let's explore the core functionalities of Power BI. What sets Power BI apart is its ability to transform raw data into compelling, insightful narratives. To make the most of this capability, it is essential to understand key features such as the Query Editor, Data Model view, visualisations and filters. These tools empower users to clean, shape, analyse and present data in a way that simplifies complexity and enhances storytelling.

After understanding the core functionalities of Power BI, we now turn our attention to the Query Editor in Power BI Desktop—an essential tool for data preparation. It enables users to clean, shape and transform raw data, laying the groundwork for accurate and meaningful analysis.

Here are the key features of the Query Editor:

- **Data connection:** Connect to various data sources, such as databases, Excel files or web services
- **Data transformations:** Perform a variety of transformations to prepare your data, including:
 - **Filtering:** Remove unnecessary rows or columns to focus on relevant data
 - **Sorting:** Arrange data in ascending or descending order for better organisation
 - **Data type conversion:** Change data types (e.g., text to number, date to datetime) to ensure consistency
 - **Adding columns:** Create new columns based on existing data to enrich your data set
 - **Merging queries:** Combine data from multiple sources into a single query for comprehensive analysis
 - **Pivoting and unpivoting:** Restructure data to facilitate better analysis and reporting

The Query Editor records each transformation as a step, allowing you to review, modify, or delete any step. This feature ensures transparency and repeatability in your data preparation process.

Another important topic to explore in Power BI's Query Editor is Applied Steps. These steps allow users to track and manage all the transformations applied to the data throughout the query process. By reviewing the Applied Steps pane, users can see each change made to the data, such as filtering, renaming columns or changing data types. This feature makes it easier to modify or remove any transformation at any stage of the data preparation process.

These applied steps are recorded in the Applied Steps pane and can be modified or deleted at any point in the query process, allowing for iterative and non-destructive data cleaning. You can also reorder steps, ensuring that transformations are applied in the correct order. This makes data preparation more transparent and flexible in Power BI.

- **Source:** This is the first step where your data source is defined. It indicates where Power BI is pulling the data from, whether it is an Excel file, a database or an online service. The Source step holds the original connection to the data before any transformations are applied.
- **Navigation:** After connecting to the data source, the Navigation step allows you to select the specific data table or sheet you want to work with. For example, if you have multiple sheets or tables in your Excel file, the Navigation step helps you choose which one to import.
- **Promoted Headers:** The Promoted Headers step helps to convert the first row of your data into column headers. This is especially useful when the data set you are working with has the column names as part of the first row of data (which might have been imported as regular data rows).

It is important to note that after loading the transformed data, you can still make further changes in the Power Query Editor. Simply open the Query Editor again by selecting 'Transform Data' from the Home tab in Power BI Desktop.

Data Modelling in Power BI

The **Data Model** view in Power BI serves as the blueprint for your data, allowing you to define relationships between tables, create calculations and set data properties.

Here are its key features:

- **Establishing relationships:** Connect tables by dragging and dropping fields to create relationships, enabling comprehensive data analysis across multiple tables.

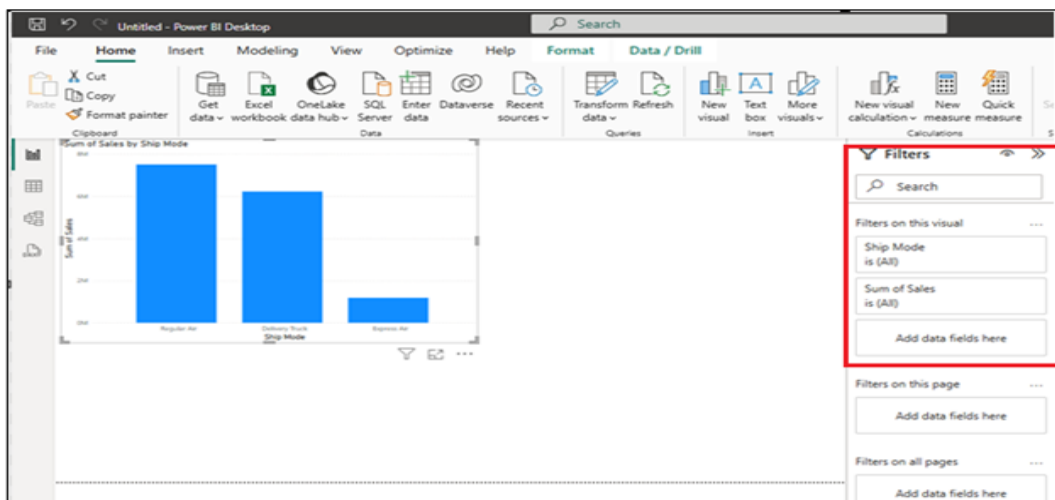
- **Creating calculated columns and measures:** Use Data Analysis Expressions (DAX) to add calculated columns and measures, facilitating custom calculations like sums, averages and percentages.
- **Defining data types and formatting:** Specify data types and formatting for each column to ensure consistency and accuracy in your reports.

A strong and accurate report in Power BI starts with building a solid foundation using the Data Model view. This feature allows you to define relationships between tables, manage fields and structure your data in a way that supports meaningful analysis. Once your data model is well-organised, you can move on to effectively communicating your insights through visualisations.

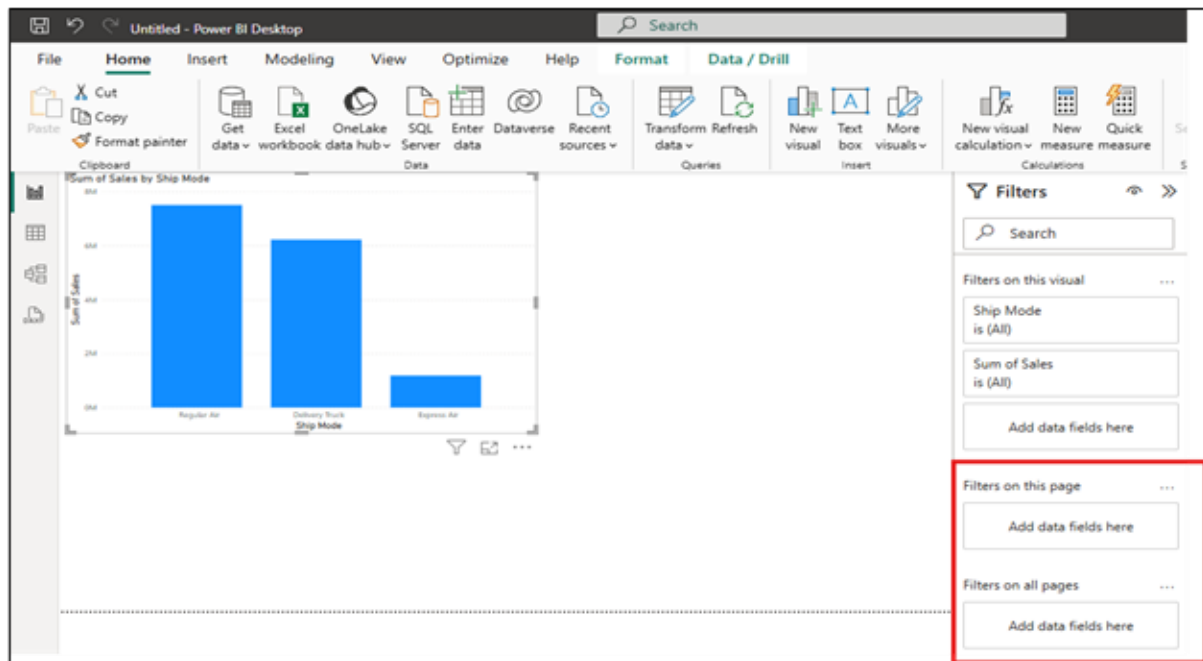
Power BI provides a wide range of built-in visuals, including bar charts, line graphs, pie charts, scatter plots and maps, which can all be customised to meet your specific reporting needs. For even more flexibility, additional visuals can be downloaded from the Power BI marketplace, allowing you to tailor your reports for maximum impact and clarity.

Another important component in Power BI is filters. Let's take a moment to understand their purpose and how they help refine and focus your data insights. **Filters** are essential for interactively exploring data by focusing on specific subsets, enabling detailed analysis and uncovering hidden patterns. The primary types of filters include:

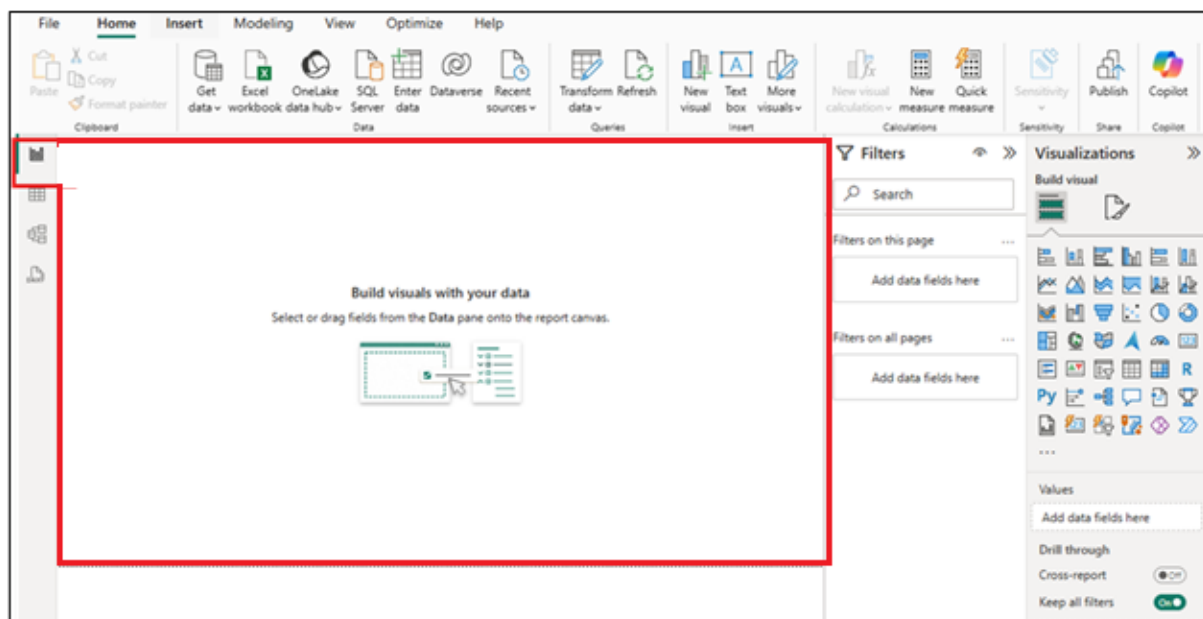
1. **Basic filters/visual-level filters:** You can apply basic filters to visualisations to display only data that meets specific criteria. These filters can be set for individual visuals, impacting only the data shown in that specific chart or graph.



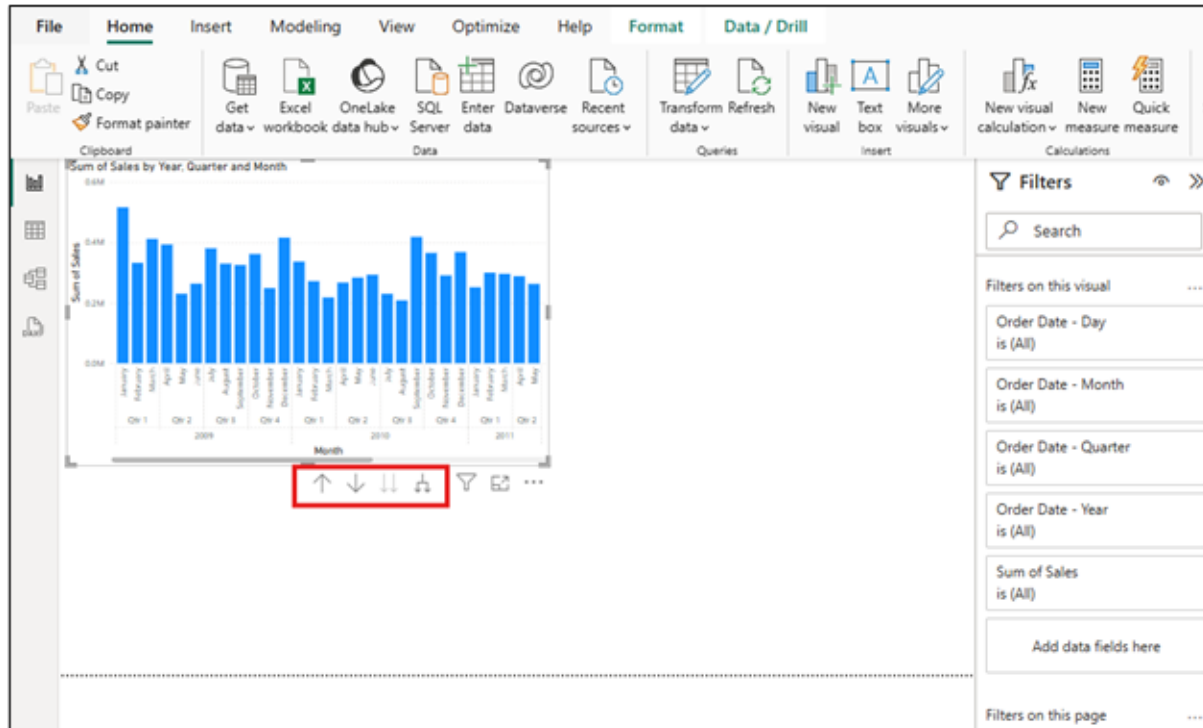
2. **Page-level filters:** These affect all visualisations on a particular report page, ensuring consistent filtering across multiple visuals within the same page.



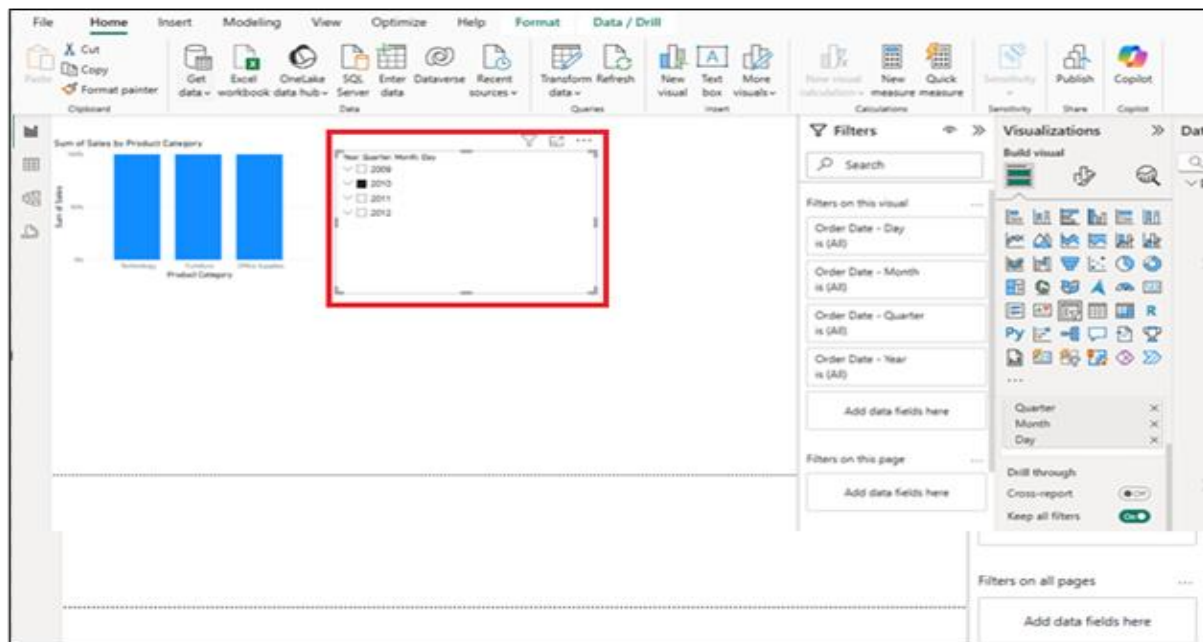
3. **Report-level filters:** These impact all visualisations across an entire report, providing a uniform filter that spans all pages and visuals within the report.



4. **Drill-through filters:** Drill-through filters in Power BI allow users to gain deeper insights by focussing on specific data points or categories. This feature enables you to create detailed reports that allow users to right-click on a data point in a visual and drill through to a new report page that is focussed on that specific data.



5. **Slicers:** These are interactive visual filters placed directly on the report canvas, allowing users to easily select and filter data by choosing values from a list or range.

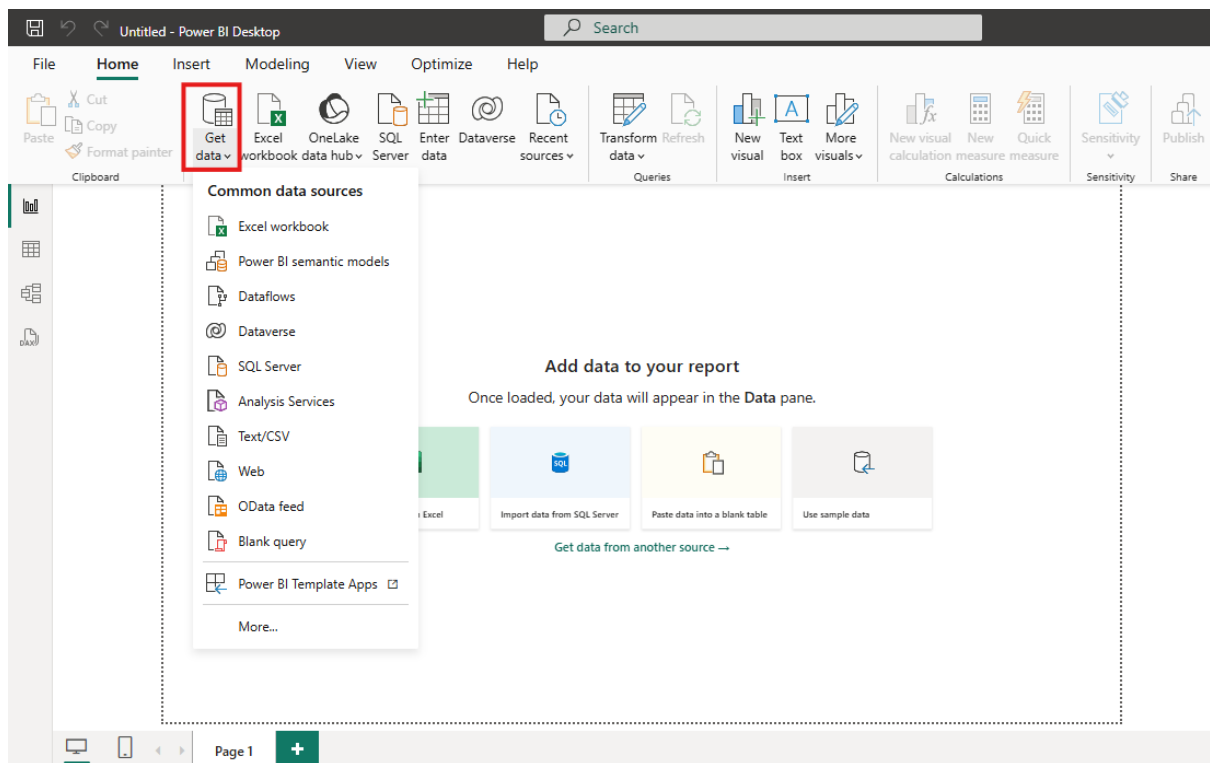


Data Connectivity and Import Options

Let's begin by exploring Power BI's powerful data connectivity and import capabilities. We will look at its flexibility in connecting to various data sources, enabling efficient data analysis and empowering better decision-making.

Power BI excels at connecting to a wide range of data sources, such as file-based, database-based and cloud-based. Whether it is Excel, SQL databases or platforms like Azure and Salesforce, Power BI enables seamless data import and transformation, helping you consolidate insights and drive data-informed decisions.

Here is a screenshot showing the different data sources available for importing data.



Let's dive into each type in detail:

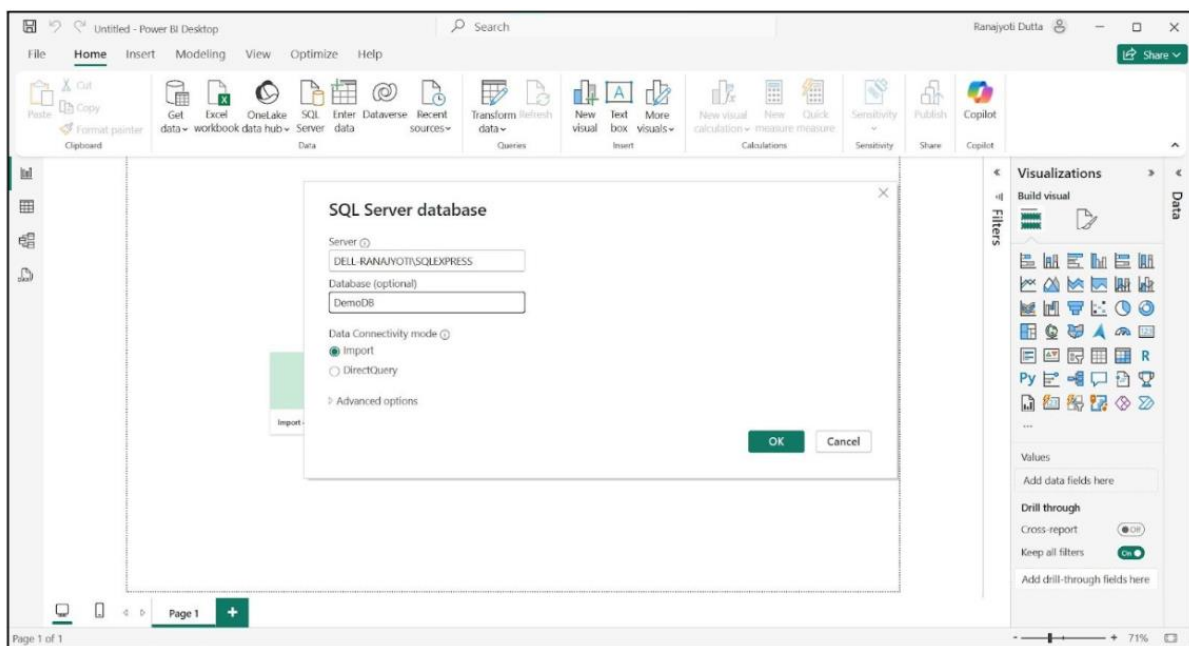
- **File-based data:** Power BI easily connects to formats like Excel, CSV, text, and JSON, making it ideal for analysing data stored locally or on shared drives.
- **Database-based data:** It supports on-premises and cloud databases such as SQL Server, Oracle, MySQL and PostgreSQL, enabling deep analysis of structured relational data.
- **Cloud-based data:** With native connectors for services like Azure, Google Analytics, Salesforce and AWS, Power BI allows seamless analysis of data from cloud platforms and SaaS applications.

Beyond standard data connections, Power BI offers flexibility to connect to custom data sources and Representational State Transfer Application Programming Interface (REST APIs) using web connectors. With Power Query, users can build API requests, handle authentication and process JSON or XML responses, making it easy to import external data. Additionally, Power BI supports parameters in queries for dynamic filtering and customisation—such as setting a date range for a sales report—enabling reusable and adaptable data imports for various use cases.

Next, let's focus on some of the most commonly used data sources for importing data into Power BI: Excel workbooks, CSV files and SQL server. After importing data, the next step is preparing it for analysis using Power Query, which offers various transformation options to structure the data for insights.

There are several options for importing data. Let's go over a few of them:

- **Excel and CSV Files:** These are ideal for importing structured data. These flat files are commonly used for reporting and can be easily transformed using Power BI's Power Query Editor. In Power BI, CSV files import simple, plain text data, while Excel files let you import data from tables or sheets with more structure and formatting. Excel gives you more options to organise your data.
- **SQL Server:** This is a robust option for connecting to relational databases. Users can choose between **Import** and **DirectQuery** modes depending on the need for performance or real-time data access (as shown in the image below).



Here are the steps for creating a new database and running queries:

1. **Open Power BI Desktop**, go to the **Home tab**, and click on 'Get Data'.
2. From the list of data sources, choose 'SQL Server'.
3. In the pop-up window, enter the **SQL Server name**. You can get this from **SQL Server Management Studio (SSMS)** if you are not sure.
4. Choose the **data connectivity mode**:
 - **Import** – if you want to bring the data into Power BI for faster performance
 - **DirectQuery** – if you want to leave the data in SQL Server and query it live
5. Click **OK** to proceed.
6. If you do not know your server name:
 - Open **SQL Server Management Studio (SSMS)**.
 - Make sure you have **SQL Server Express** and **Microsoft Access Database Engine** installed.

- In the connection window, you will see the **Server Name**. Tick the '**Trust server certificate**' option and click Connect.
7. To create a test database:
- In SSMS, right-click '**Databases**', then select '**New Database**'.
 - Name it **DemoDB** and click OK.
 - Then, right-click on **DemoDB**, choose '**New Query**', and use SQL to add data.
8. Once connected in Power BI, you will see a list of tables and views. Select the ones you need and load them into Power BI.

Here is a quick example query:

```
USE DemoDB;
```

```
CREATE TABLE Person (
```

```
    ID INT PRIMARY KEY,
```

```
    Name NVARCHAR(50),
```

```
    Age INT
```

```
);
```

```
INSERT INTO Person (ID, Name, Age)
```

```
VALUES
```

```
    (1, 'John Doe', 30),
```

```
    (2, 'Jane Smith', 25),
```

```
    (3, 'Mike Johnson', 40);
```

Here are some additional ways to import data that you can explore:

- **SharePoint Folder:** This is useful for importing files (like Excel or CSV) stored in SharePoint document libraries. This is especially beneficial for collaborative environments.
- **Web:** Power BI can connect to publicly available data or APIs through a URL, enabling integration with online data feeds or web-based data sets.
- **Azure Services:** Power BI integrates with several Azure services, such as Azure SQL Database, Azure Data Lake and Azure Blob Storage, allowing for scalable and cloud-based data analysis.

- **OData Feed:** This is a protocol that allows access to data from a variety of services, commonly used in enterprise applications like Dynamics 365.
- **Text/JSON/XML Files:** For working with semi-structured data formats, Power BI can import and transform these file types effectively.
- **Google Analytics:** Built-in connectors allow users to pull web analytics data directly from their Google Analytics account.
- **Power BI Data Sets and Dataflows:** This enables the reuse of already modelled data within your Power BI environment for consistency and efficiency across reports.

Power BI allows data to be brought in from SQL Server using two connectivity modes. These are the two available options for importing data:

DirectQuery mode

- Data stays in the source; no data is imported into Power BI.
- Every interaction triggers a live query to the source (e.g., SQL database).
- It always shows up-to-date data.
- It may be slower due to constant querying.

Import mode

- A snapshot of the data is imported and stored in Power BI.
- Interactions use cached data for faster performance.
- It is ideal for smaller data sets (<1 GB) that do not change frequently.
- It requires manual or scheduled refresh for updates.

The following table outlines how different data connectivity modes compare based on key parameters.

Parameter	Import Mode	DirectQuery Mode
Refresh Frequency	Hourly or daily	Real-time
Performance	As the data model is already cached, there is no visible latency.	Queries are processed in real time and depend on network speed and source system performance.
Data Storage	Data is stored in Power BI Service (cloud).	Power BI Service does not store data; it remains in the on-premises source.
Max. Data Set Size	It is limited to 1 GB per model (higher limits are available for Premium users).	There is no storage limit, as data resides in the on-premises database.

Datasource Support	It supports multiple data sources.	It supports only one data source and allows filtering in a single direction only.
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Data Preparation

Data preparation and cleaning are essential steps in the data analysis process. Data preparation involves organising and transforming raw data into a usable format, while data cleaning focusses on identifying and correcting errors, removing duplicates and handling missing or inconsistent values. Together, these processes ensure that the data is accurate, reliable and ready for analysis or reporting.

Data Cleaning

Data cleaning begins by importing a data set and selecting all three tabs. After clicking on 'Transform Data', the Power Query editor opens. You will observe that the column headers in the 'Returns' and 'People' tables are incorrectly placed in the first row. To fix this, go to the 'Home' tab and click on 'Use First Row as Headers' to properly assign the column names.

Performing a Simple Extract

In Power BI, extracting specific parts of data from existing columns can help simplify your analysis. For instance, as demonstrated in the video, if you need to extract the year from an 'Order ID' column where the year is enclosed by hyphens, you can do so without altering the original data. This can be efficiently accomplished using the 'Text Between Delimiters' option in Power Query.

This task highlights a common data transformation technique used in Power BI: extracting specific information from a text field without altering the original data. By creating a new column and using text extraction tools (like 'Text Between Delimiters'), users can isolate meaningful values—such as dates, codes or identifiers—embedded within larger text strings. This method ensures the original data remains untouched while enhancing its usefulness for analysis and reporting.

Additional tasks that can be performed using the Power Query Editor include:

Pivot and Unpivot Data:

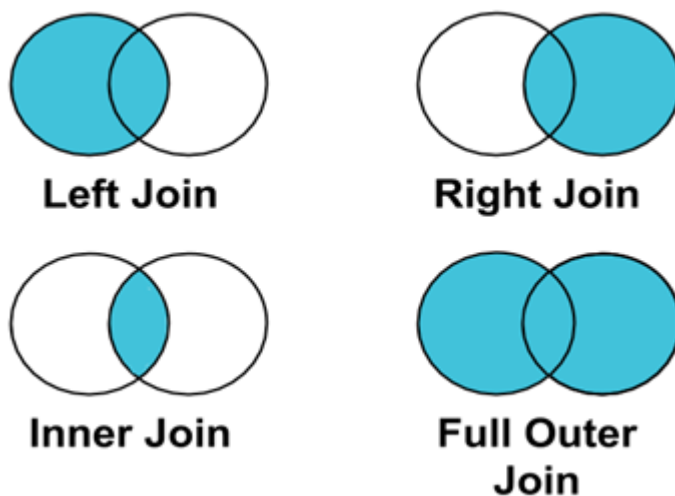
- Unpivoting helps transform cross-tab or wide-format data into a structured, column-based format that is easier to analyse.
- Pivoting allows you to group and summarise data by specific categories, making it more suitable for reporting and visualisations.

Power BI provides two key features in Power Query for combining data from multiple tables—**merge** and **append**. Merging is used to join tables based on common columns, allowing related information to be brought together into a single table. Appending is used to stack tables with similar structure one below the other, effectively combining their rows. These features are useful when working with fragmented data sets that need to be unified for analysis.

There are four main types of joins:

1. **Inner join:** Returns only the rows that are common to both tables, similar to the intersection of two sets ($A \cap B$)
2. **Left outer join:** Returns all rows from the left table and any matching rows from the right table
3. **Right outer join:** The reverse of the left outer join, returning all rows from the right table and matching rows from the left
4. **Full outer join:** Combines all rows from both tables, similar to the union of two sets ($A \cup B$)

Below is a visual representation of the different join types.



In addition, there are **left-anti** and **right-anti joins**. The left-anti join returns rows from the left table that have no matching rows in the right table, while the right-anti join does the reverse.

Let's dive deeper into some other important concepts related to Power BI.

1. Fuzzy matching

- Fuzzy matching helps match text values that are similar but not exactly the same.
- It is used when merging tables with slight differences in spelling or formatting (e.g., 'Jon Smith' vs 'John Smith').
- You can adjust the similarity threshold to control how close two values need to be to match.
- It is useful when working with inconsistent or messy data, especially from different sources.

2. Splitting columns using Power Query

As demonstrated in the video, you learnt how to use Power Query to split columns and saw how this feature plays a key role in transforming and organising data effectively.

- In this example, we are using Power BI's 'Split Column' feature to separate full customer names into first and last names.
- To do this, go to the 'Home' tab, click 'Split Column' and choose 'By Delimiter'. Since names are separated by spaces, select 'Space' as the delimiter. Then, choose the 'left-most delimiter' option to split at the first space.
- Splitting columns in Power BI helps break down combined data into separate parts, making it easier to analyse, filter and organise. It improves data clarity and supports more effective reporting and modelling.

3. **Appending queries in Power Query:** Let's explore the Append Queries feature. It helps combine data from multiple tables with the same structure into one unified table. This is useful when you have data sets with the same columns (e.g., sales data from different months) and want to consolidate them for analysis. By appending queries, you can seamlessly merge data from various sources or periods into one cohesive data set.

Data Modelling

Data modelling is a key step in structuring data effectively within Power BI. It involves creating relationships between tables, organising data into fact and dimension tables and leveraging the

data model to enhance the quality of analysis and reporting. A well-structured model forms the foundation for building insightful and efficient reports.

Data modelling in Power BI involves several important concepts that help structure and optimise data for analysis:

- **Establishing relationships between tables:** Linking tables through common fields allows accurate data analysis.
- **Building calculated columns and measures with DAX:** DAX creates new columns or dynamic measures for more advanced calculations.
- **Creating hierarchies and keys:** Hierarchies organize data for easier navigation, while keys ensure data integrity.
- **Improving performance by reducing redundancy:** Removing duplicates and unnecessary data enhances query speed and memory efficiency.

A common and effective approach to data modelling is the **star schema**, where a central fact table containing quantitative data is connected to surrounding dimension tables that provide context. This structure is simple and improves the performance of Power BI reports.

Another important concept is **cardinality**, which defines how tables relate to each other within a data model. Cardinality types include:

- **One to many:** One record in one table relates to multiple records in another.
- **One to one:** One record in one table matches exactly one record in another.
- **Many to one:** Multiple records in one table link back to a single record in another.
- **Many to many:** Records in both tables can relate to multiple records in the other.

Data modelling helps define meaningful connections between tables, such as linking a Product ID in the Customers table to its corresponding name in the Product table. Tables are generally categorised as fact tables (e.g., Customer, Recovery) and dimension tables (e.g., Branch, Product).

For better organisation, place dimension tables at the top and fact tables below. To delete a relationship, right-click and select Delete. To create a relationship, drag a common field (like Product ID) from one table to the matching field in another. A successful relationship typically shows a one-to-many link, active status and single cross-filter direction. Do not forget to click Save to confirm.

When creating relationships in Power BI, ensure both fields have the same data type—either text or numeric—or the relationship will not work. Avoid circular relationships, as they cause errors. Resolve them by deleting or deactivating one link (active links show as solid lines;

inactive as dotted). Use Manage Relationships > Autodetect to let Power BI auto-create valid connections based on matching columns.

Power BI typically detects these relationships automatically when data is imported. However, they may not always be accurate. If necessary, you can adjust the relationships in the model view to meet your specific needs.

Next, let's understand how cross-filtering works. Cross-filtering enables interactions between visualisations—like charts and tables—within a report, allowing data in one visual to filter or highlight related data in another.

For example, if you have a sales report with a bar chart showing sales by product and a table displaying sales by region, selecting a specific product (e.g., Product A) in the bar chart will automatically update the table to show sales for Product A in each region. This is made possible by cross-filtering, which links visual elements so that interactions in one chart affect others.

There are two types of cross-filtering:

1. **Single-direction filtering:** This is where filtering happens in only one direction. For example, selecting a product filters the sales table.
2. **Bi-directional (both) filtering:** This is where filtering happens both ways. For example, selecting a product updates the sales table, and selecting a region in the table updates the product chart.

Data Visualisation using Power BI

Let's begin our exploration of Power BI visualisations by covering core chart types like stacked bar charts, line charts, and tree maps.

- **Stacked bar chart:** In a stacked bar chart, data is represented horizontally, with segments stacked within each bar. It shows the total and the contribution of each part within a category.
- **Line chart:** Line charts are ideal for visualising trends over time. They connect data points with lines, making it easy to observe patterns or changes in data across dates or categories.
- **Tree maps:** They are a type of visual in Power BI that display hierarchical data using nested rectangles. Each rectangle represents a category, and the size and colour of the rectangles are used to indicate different metrics (like total sales, profit, etc.).

- **Map visual:** This visual places data on a geographic map, helping users analyse location-based metrics like regional sales or customer distribution.
- **Pie chart:** A pie chart breaks down data into slices to show proportions of a whole. It is useful when you want to show how individual segments contribute to a total.
- **Donut chart:** This is similar to a pie chart but with a circular cut-out in the centre; the donut chart offers a cleaner aesthetic while also representing parts of a whole.
- **Card visual:** The card visual highlights a single numeric value, such as a key performance indicator (KPI). It is perfect for showing metrics like total sales, profit or counts at a glance.
- **Matrix:** A matrix works like a pivot table, displaying data in rows and columns. It allows for grouping and summarising data by multiple dimensions, which is helpful for detailed data analysis.
- **Table:** Tables provide a detailed view of data in its raw format. They are effective when exact values or multiple fields need to be displayed together.

Each visualisation in Power BI serves a unique purpose, whether it is to compare, track trends, display proportions or filter data. By choosing the right visual, users can make their reports more insightful, interactive and easier to interpret, turning raw data into compelling stories and better decisions.

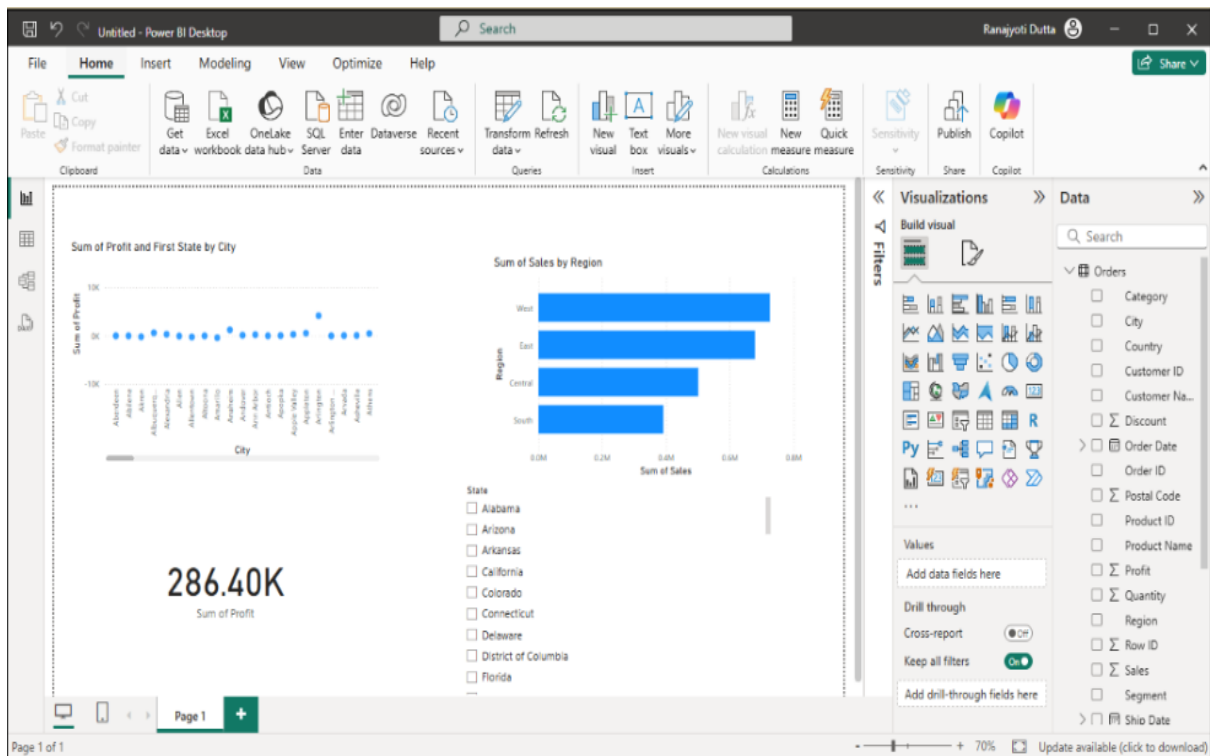
Let's take a brief look at custom visuals in Power BI. Custom visuals are user-defined or third-party visuals that go beyond the default set provided by Power BI. They allow users to extend the platform's visualisation capabilities to better suit specific business needs or aesthetic preferences.

- **Sankey chart:** For visualising flow or movement across categories.
- **Bullet Chart:** For comparing performance against a target.
- **KPI Ticker:** For scrolling KPIs.
- **Chiclet Slicer:** For a button-style slicer with images or custom formatting.

Report Generation using Power BI

This involves using visuals like charts, tables and slicers to present key insights. The layout and interactions are then customised to ensure the report is interactive, intuitive and user-friendly.

1. **Graph 1:** Sum of Profit by City, with State added as tooltips (scatterplot)
2. **Graph 2:** Sum of Sales by Region (bar chart)
3. **Graph 3:** Sum of Profit (card visual)
4. **Graph 4:** Slicer visual for State



5. When creating a report, there are several key considerations to keep in mind:

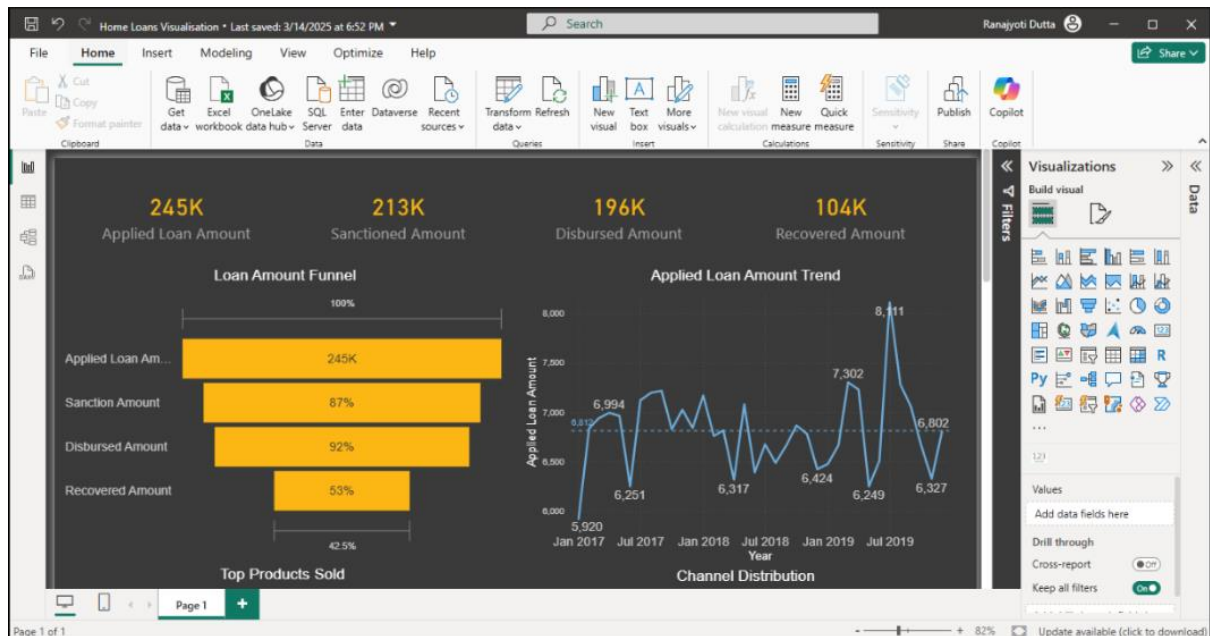
- **Consistency:** Maintain a professional look by using a uniform colour scheme, font size and alignment.
- **Navigation:** Enhance user experience by integrating slicers, drill-through and buttons for seamless navigation.
- **Data Efficiency:** optimise performance by removing unnecessary fields and limiting row-level details to reduce data load.
- **Mobile optimisation:** If required, optimise the report layout for mobile devices. To optimise for mobile, simply go to the 'Mobile' tab, where you can either manually adjust the visuals or use the 'auto-create' options.

Following are the steps in saving a Report in Power BI

- Save your work in Power BI Desktop as a .pbix file, which contains visuals, data connections and DAX formulas.
- Publish the report to Power BI Service for online access, viewing and collaboration.
- Enable sharing and collaboration through the Power BI Service for easier access and updates.
- Easily schedule refreshes and manage reports on the cloud platform.

For example, in the home loans report [here](#), we can see a well-organised design with key metrics displayed at the top as shown in the image below. The colour scheme is consistent, and there are interactive filters for user engagement. The visuals include an ArcGIS chart for branch

distribution, a pie chart and a funnel chart. Additionally, the report offers drill-down options for approved loan amounts and features buttons that allow users to toggle between the decomposition tree and Q&A visuals.



Benefits of using Copilot for Report Generation

Here are some benefits of using Copilot for report creation in Power BI:

- **Natural language input:** You can describe the report you want in plain language (e.g., 'sales performance by region').
- **Automatic report generation:** Copilot reads your instructions and generates a report with relevant visuals based on your data set.
- **Customisable output:** You can ask Copilot to emphasise trends or add specific elements like charts or filters.
- **Time-saving:** This approach speeds up the report creation process, especially useful for users with less experience in building visuals.
- **Interactive and flexible:** Copilot supports ongoing refinement—you can keep giving instructions to tweak or improve the report.

You can guide Copilot to recreate reports and emphasise key trends. It can also generate a summary of the report, including clickable references that link to the specific visuals or sections used to create the summary.

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